



Leaders In An Array

An element is leader, if it is greater than all the elements to its right side. And the rightmost element is always a leader.



Approaches to find the Leaders In An Array



Brute Force Algorithm

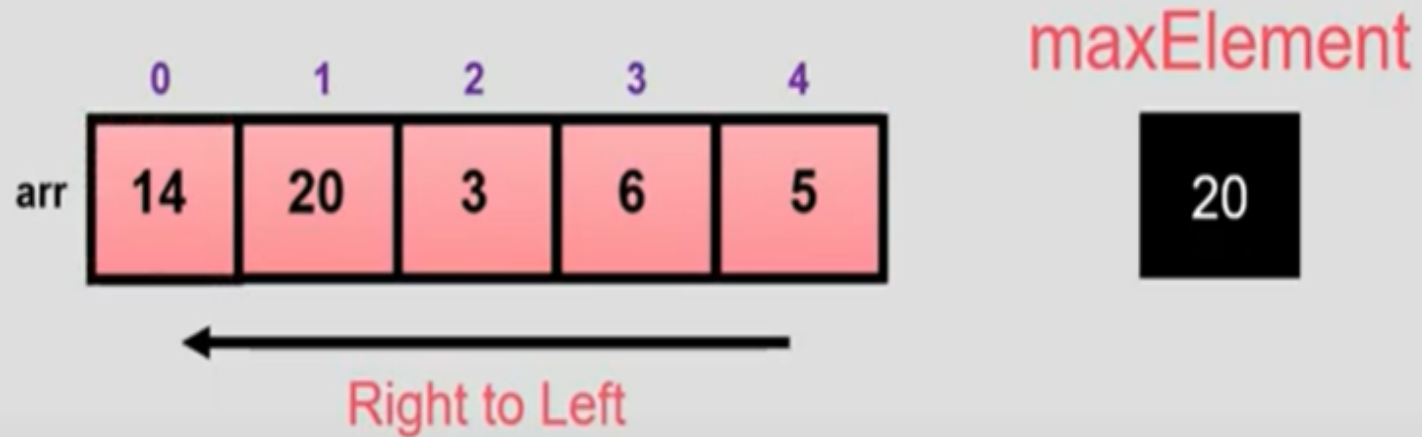


Leaders are: 20, 6, 5

Brute Force Algorithm

- Time complexity is $O(n * n)$
- In this approach we use two loops
- Outer loop is use to traverse the array from 0 to $n-1$
- Pick one by one all the elements of the array from left to right
- Then inner loop is use to compares the picked element with all the elements to its right side

Scan Array from Right



Leaders are: 5, 6, 20

Scan Array from Right

- Time Complexity is: $O(n)$
- In this approach, we traverse from right side to left side($n-1$ to 0) of the array.
- Keep the one highest variable in it that is use as a highest variable
- If we find any element that is greater than this highest variable then we print this element as a leader
- And update the highest variable with the new value

Brute Force Algorithm

```
public class LeadersInArray {  
    public static void main(String[] args) {  
        int[] array = {16, 17, 4, 3, 5, 2};  
        findLeaders(array);  
    }  
  
    public static void findLeaders(int[] array) {  
        int length = array.length;  
  
        for (int i = 0; i < length; i++) {  
            boolean isLeader = true;  
  
            for (int j = i + 1; j < length; j++) {  
                if (array[i] <= array[j]) {  
                    isLeader = false;  
                    break;  
                }  
            }  
        }  
    }  
}
```

```
        if (isLeader) {  
            System.out.print(array[i] + " ");  
        }  
    }  
}
```


Scan Array from Right

```
public class LeadersInArray {  
    public static void main(String[] args) {  
        int[] array = {16, 17, 4, 3, 5, 2};  
        findLeaders(array);  
    }  
  
    public static void findLeaders(int[] array) {  
        int length = array.length;  
        int max = array[length - 1]; // The last element is always a leader  
        System.out.print(max + " ");  
  
        for (int i = length - 2; i >= 0; i--) {  
            if (array[i] > max) {  
                max = array[i];  
                System.out.print(max + " ");  
            }  
        }  
    }  
}
```



THANK YOU